

Keeper tier-gate — Lyra Spectral Score Theoretical Layer v0.1 (Movie V0 backbone) + Vol 16 Ch1 §3 v0.1 ($A_{\text{sub}} \leftrightarrow U_q(B_2)$ dictionary at operator level). Both PASS at FRAMEWORK / PEDAGOGICAL tier; neither introduces new derivations. Spectral Score: gluon placement consistency note; Higgs $V_{(0,0)}$ provisional pending L5. Vol 16 §3: color 8-vs-3 gap to mark explicit; 15 vs 26 generator count clarification.

Keeper

2026-05-30 Saturday

Tier-gate — Spectral Score v0.1 + Vol 16 Ch1 §3 v0.1

1. Lyra Spectral Score Theoretical Layer v0.1 (Movie V0 backbone)

Verdict: PASS at FRAMEWORK / ORGANIZATIONAL. The doc is the theoretical layer for the V0 visualization artifact, not a derivation. Organizes existing tier-closed content (K-type Casimirs, substrate-primary anchors, T190/T2003/T187 ratios, F1 PMNS, quasi-eigentone decay arrows, Five-Absence gaps). Honest scoping: “this v0.1 is the THEORETICAL layer for a visualization artifact — it doesn’t derive new content but ORGANIZES the dictionary’s existing placements.”

Strengths: - Sector layout coherent (Higgs, lepton-Shilov, vector-gauge, adjoint-gauge, bulk-fundamental, higher composites). - Substrate-primary anchor table is a clean visualization-ready summary (Casimirs 0, 5/2, 4, 6, 10, 15/2, 21/2 mapped to roles). - Mass-spacing references properly attributed to T190, T2003, T187, Weinberg derivations, F1 PMNS. - Decay arrow table absorbs the quasi-eigentone framework cleanly. - Five-Absence visualization (no 4th gen, no SUSY, no GUT leptoquarks, no $0\nu\beta\beta$, no monopole) is the right gap-visualization framing. - Movie V0 narrative spine is structurally sound.

Notes for v0.2 (Cal cold-read recommended):

- N1 — Gluon K-type placement consistency: Section 1 lists gluon at $V_{(1,0)}$ (vector-gauge, Casimir 4) AND at $V_{(1,1)}$ (adjoint-gauge, “gluon-adjoint”) AND implicitly at bulk $SO(3)$ sub-vector per bulk-color v0.4. Three potential placements coexist in the same doc. Suggested v0.2 resolution: mark gluon as OPEN pending bulk-color (C+D) closure; visualize provisionally at $V_{(1,1)}$ adjoint (Casimir 6 = C_2) per the gauge-adjoint pattern, with explicit caveat. The substrate-level gluon placement is multi-week per bulk-color v0.4.
- N2 — Higgs at $V_{(0,0)}$ provisional: Section 1 places Higgs at “ $V_{(0,0)}$ trivial, Casimir 0.” This is provisional pending L5 absolute scale + Higgs/EWSB substrate-derivation (A3 macro program, OPEN). For Movie V0, mark Higgs placement as provisional pending L5 closure; the scalar nature is structurally correct (Higgs is the BEH scalar) but the precise K-type identification + relation to vacuum is OPEN.

Tier disposition: - Theoretical organization: FRAMEWORK (not derivation, but coherent visualization spine) - K-type Casimir anchors: RIGOROUS (substrate-primary values verified) - Mass-spacing references (T190, T2003, T187, Weinberg, F1): RIGOROUS or D-tier (carried from cited theorems) - Decay arrows: FRAMEWORK (from quasi-eigentone v0.1, already tier-gated) - Five-Absence gaps: FRAMEWORK (F5 already STANDING) - Gluon

placement, Higgs placement: OPEN, provisional (N1, N2)

Status: PASS as Movie V0 organizational layer; ready for Grace/Elie artifact build with N1+N2 caveats marked explicitly in the visualization.

2. Lyra Vol 16 Ch1 §3 v0.1 ($A_{\text{sub}} \leftrightarrow U_q(B_2)$ dictionary at operator level)

Verdict: PASS at PEDAGOGICAL SYNTHESIS. Updates the genus convention to the Thursday-2026-05-28 corrected one-genus + Casimir + embedding (per A1 v0.5 closure). Organizes the Saturday dictionary work (L1-L3, engine v0.1-v0.4, quasi-eigentone, 5-tuple taxonomy) into Vol 16's operator-algebraic spine. Connects to Vol 0/1/2/10.

Strengths: - Genus convention update (Section 0) per A1 v0.5 — correctly states $\text{GENUS} = n_C = 5 \neq C_2 = 6 = \text{Casimir} \neq g = 7 = \text{embedding}$. Calibration #33 sourcing: “Faraut-Korányi 1994 cited at footnote level” is acceptable for pedagogical use given values independently derived/established in BST work. - $A_{\text{sub}} \leftrightarrow$ Hall algebra dictionary table (Section 1.1) organizes the operator-K-type-Hall-image triple cleanly per generator family. - Engine structure connection (Section 1.2) anchors to the established Serre coefficients ($[2]_2=3=n_C$ short; $[3]_4=21=n_C \cdot g$ long) and Drinfeld pairing numerators ($N_c, N_c \cdot n_C$). - 5-tuple taxonomy (Section 2) at operator level is consistent with L1-L3 derivations. - Quasi-eigentone framework (Section 3) at operator level is the right pedagogical translation. - Cross-volume connection (Section 4) places Vol 16 as the bridge from Vol 0 + Vol 10 to Vol 1 + Vol 2.

Notes for v0.2:

- N1 — Color $\hat{C}_3$ (8 generators) vs bulk $SO(3)$ sub-vector (3 generators): Section 1.1 lists “Color $\hat{C}_3$ (8 generators) $\leftarrow V_{(1,1)}$ bulk $SO(3)$ sub-vector $\rightarrow$ BULK-COLOR (Family 4).” The 8-vs-3 dimensional gap is the load-bearing multi-week burden per bulk-color v0.4: $SO(3)$ has adjoint dim 3, $SU(3)$ (QCD) has adjoint dim 8, and the bulk-color (C+D) commitment hypothesizes the 8 emerges from Toeplitz algebra on $H^2(S)$ acting on the 3 $SO(3)$ labels. v0.2 should mark “Color $\hat{C}_3$ (8 generators)” with $OPEN \oplus$ pending bulk-color (C+D) closure; without that caveat the table understates the gap.
- N2 — 15 vs 26 generator count clarification: Section 1.1 references “15 A_{sub} generators per A_{sub} v0.12 enumeration.” The table itself lists ~26 distinct operator entries (Hamiltonian + 5 position + 5 momentum + 3 spin + 3 angular + Bell + T + C + P + N + γ^5 + Q + 8 color + I_3 + σ_{BF} = ~31 if all components count). The “15” must be the family-count (position-family as 1 generator type, etc.). Suggest v0.2 clarify “15 generator families, with component-level expansion to ~26-31 distinct operators.”

Tier disposition: - Genus convention update (Section 0): RATIFIED (per A1 v0.5 + Thursday Cal #30/#31 STAND-ING) - $A_{\text{sub}} \leftrightarrow$ Hall algebra dictionary (Section 1.1): PEDAGOGICAL SYNTHESIS (organizes existing L1-L3 + engine content) - Engine structure constants (Section 1.2): RIGOROUS (E_1/E_2 /Serre/Drinfeld from established toys) - 5-tuple taxonomy at operator level (Section 2): FRAMEWORK (consistent with L1-L3) - Quasi-eigentone at operator level (Section 3): FRAMEWORK (consistent with quasi-eigentone v0.1) - Cross-volume connection (Section 4): ORGANIZATIONAL

Status: PASS as Vol 16 Chapter 1 §3 pedagogical content; N1+N2 sharpening for v0.2; Vol 16 spine substantively coherent with A1 v0.5 + Saturday dictionary work.

3. Routed handback

- Lyra: Spectral Score v0.2 absorbs N1 (gluon placement open marker) + N2 (Higgs $V_{(0,0)}$ provisional marker). Vol 16 Ch1 §3 v0.2 absorbs N1 (color 8-vs-3 open marker) + N2 (15-vs-26 generator-count clarification).
- Grace: V0 spectral score artifact build can proceed using v0.1 backbone; mark gluon and Higgs placements per N1+N2 caveats in the visualization (e.g., dotted/dashed rendering for open placements).
- Elie: when engine v0.3 V-compound layer + bulk-color (C+D) toy-level investigation lands, both Spectral Score and Vol 16 §3 consolidate.
- Cal: cold-read queue extends with Spectral Score v0.1 + Vol 16 §3 v0.1 (both PEDAGOGICAL SYNTHESIS; structural-coherence cold-read appropriate).

4. Queue continuation

Saturday afternoon tier-gates total: 7 items dispositioned.

1. Bulk-color v0.4 — FRAMEWORK with C1+C2
2. Engine v0.2 audit support v0.2 — PASS; §6+§7 STRUCTURAL pre-reserved
3. T1 falsifier v0.1 — PASS with v0.2 sharpening
4. Two-Route Scan v0.1 — STRUCTURAL with N1+N2 (null-model + cluster wording)
5. Quasi-Eigentone Framework v0.1 — FRAMEWORK with N1+N2 (ν -eigenstate + V-compound)
6. Spectral Score Theoretical Layer v0.1 — FRAMEWORK with N1+N2 (gluon + Higgs)
7. Vol 16 Ch1 §3 v0.1 — PEDAGOGICAL with N1+N2 (color 8-vs-3 + generator count)

Remaining in immediate queue: - Engine v0.3 consolidation doc RE-AUDIT (pending Elie filing; the audit-support doc was preparatory) - Casey-Saturday's other landed items: Grace Periodic Table v0.7, Grace G12 Hadron v0.2, Lyra Strong-Uniqueness v1.1, Grace Mendeleev v0.1, Calibration #34 candidate, Grace G13 8-gluon, Cal Proposal K-types-are-particles falsifier design v0.1

Pulling next: Calibration #34 candidate (Cal pattern-watch convergence) — 3-strike count check from this week; high-leverage methodological item.

— Keeper. Two more tier-gates filed (Saturday afternoon batch now 7); both pedagogical-synthesis items, neither load-bearing for new derivations; team queue continues; sustainable cadence held.